

Indian Statistical Institute, Bangalore

B. Math.

First Year, Second Semester

Real Analysis-II

Back Paper Examination

Maximum marks: 100

June 3, 2024

Time: 3 hours

Instructor: B V Rajarama Bhat

In the following a, b are real numbers with $a < b$.

- (1) Suppose $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ are bounded Riemann integrable functions. Show that fg defined by $fg(x) = f(x)g(x)$, $\forall x \in [a, b]$ is Riemann integrable. [15]
- (2) Give an example of two bounded real valued functions h, k on $[a, b]$ such that h, k are not Riemann integrable but both $h + k$ and hk are Riemann integrable. [15]
- (3) Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Show that there exists $c \in [a, b]$ such that

$$\int_a^b f(t)dt = f(c)(b - a).$$

[15]

- (4) Explicitly construct a sequence of continuous functions $\{f_n\}_{n \geq 1}$ converging pointwise to the function $h : [-1, 1] \rightarrow \mathbb{R}$ defined by

$$h(x) = \begin{cases} 1 & \text{if } -1 \leq x < 0; \\ 0 & \text{if } x = 0; \\ -1 & \text{if } 0 < x \leq 1. \end{cases}$$

Show that the convergence is not uniform.

[15]

- (5) Let $\{f_n\}_{n \geq 1}$ be a sequence of continuous functions on $[0, 1]$ converging pointwise to a continuous function f on $[0, 1]$. Suppose $f_n(x) \geq f_{n+1}(x)$, $\forall n \geq 1$ and $0 \leq x \leq 1$. Show that $\{f_n\}_{n \geq 1}$ converges uniformly to f . [15]
- (6) Let $g : [0, 2] \rightarrow \mathbb{R}$ be a function defined by

$$g(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1; \\ 1 & \text{if } 1 < x \leq 2. \end{cases}$$

Obtain a sequence of polynomials converging uniformly to g on $[0, 2]$.

[15]

- (7) Consider a power series $\sum_{k=0}^{\infty} a_k x^k$, $a_k \in \mathbb{R}$. Show that if $\{|a_k x_0^k| : 0 \leq k < \infty\}$ is bounded for some $x_0 \in \mathbb{R}$, then $\sum_{k=0}^{\infty} a_k x^k$ is absolutely convergent for $0 \leq |x| < |x_0|$. [15]
- (8) Suppose p is a real polynomial. Show that the Taylor series of p around any point $a \in \mathbb{R}$ is the polynomial itself. [15]